

5.9 Classwork: Differentiation and Integration Drill — Markscheme

— **Differentiation** — $f(x) = 4x^3 - 6x + 7$ $f'(x) = 12x^2 - 6$ (A1) $y = 5 \sin x - 2 \cos x$
 $\frac{dy}{dx} = 5 \cos x + 2 \sin x$ (A1)(A1) $f(x) = 3e^{2x}$ $f'(x) = 3 \cdot 2e^{2x} = 6e^{2x}$ (M1)(A1)
 $y = 4 \ln x + 3x^{-1}$ $\frac{dy}{dx} = \frac{4}{x} - \frac{3}{x^2}$ (A1)(A1) $f(x) = \sin(3x + 1)$ *chain rule*
 $f'(x) = 3 \cos(3x + 1)$ (M1)(A1) $y = e^{x^2-1}$ *chain rule* $\frac{dy}{dx} = 2x \cdot e^{x^2-1}$ (M1)(A1)
 $f(x) = x^2 \ln x$ *product rule* $f'(x) = 2x \cdot \ln x + x^2 \cdot \frac{1}{x} = 2x \ln x + x = x(2 \ln x + 1)$
(M1)(A1)(A1) — **Integration** — $\int (6x^2 + 4x - 1) dx = 2x^3 + 2x^2 - x + c$ (A1)(A1)
 $\int \left(3 \cos x + \frac{1}{x} \right) dx = 3 \sin x + \ln x + c \quad (x > 0)$ (A1)(A1) $\int 5e^{3x} dx = \frac{5}{3}e^{3x} + c$
(M1)(A1) $\int_0^\pi 2 \sin x dx = [-2 \cos x]_0^\pi$ (M1)
 $= -2 \cos \pi - (-2 \cos 0) = -2(-1) + 2(1) = 4$ (A1)(A1)
 $\int_1^2 \left(\frac{3}{x} + 2x \right) dx = [3 \ln x + x^2]_1^2$ (M1) $= (3 \ln 2 + 4) - (0 + 1) = 3 \ln 2 + 3$ (A1)(A1)
 $f'(x) = 4x + e^x$, $f(0) = 3$ $f(x) = 2x^2 + e^x + c$ (A1) $f(0) = 0 + 1 + c = 3 \Rightarrow c = 2$
(M1) $f(x) = 2x^2 + e^x + 2$ (A1) $f(x) = (2x + 1)e^x$ $f'(x) = 2e^x + (2x + 1)e^x = (2x + 3)e^x$
(M1)(A1) Hence $\int (2x + 3)e^x dx = (2x + 1)e^x + c$ (M1)(A1)